Cloud to Code: Using Google Earth Engine, R, and AI to Streamline Geospatial Analysis

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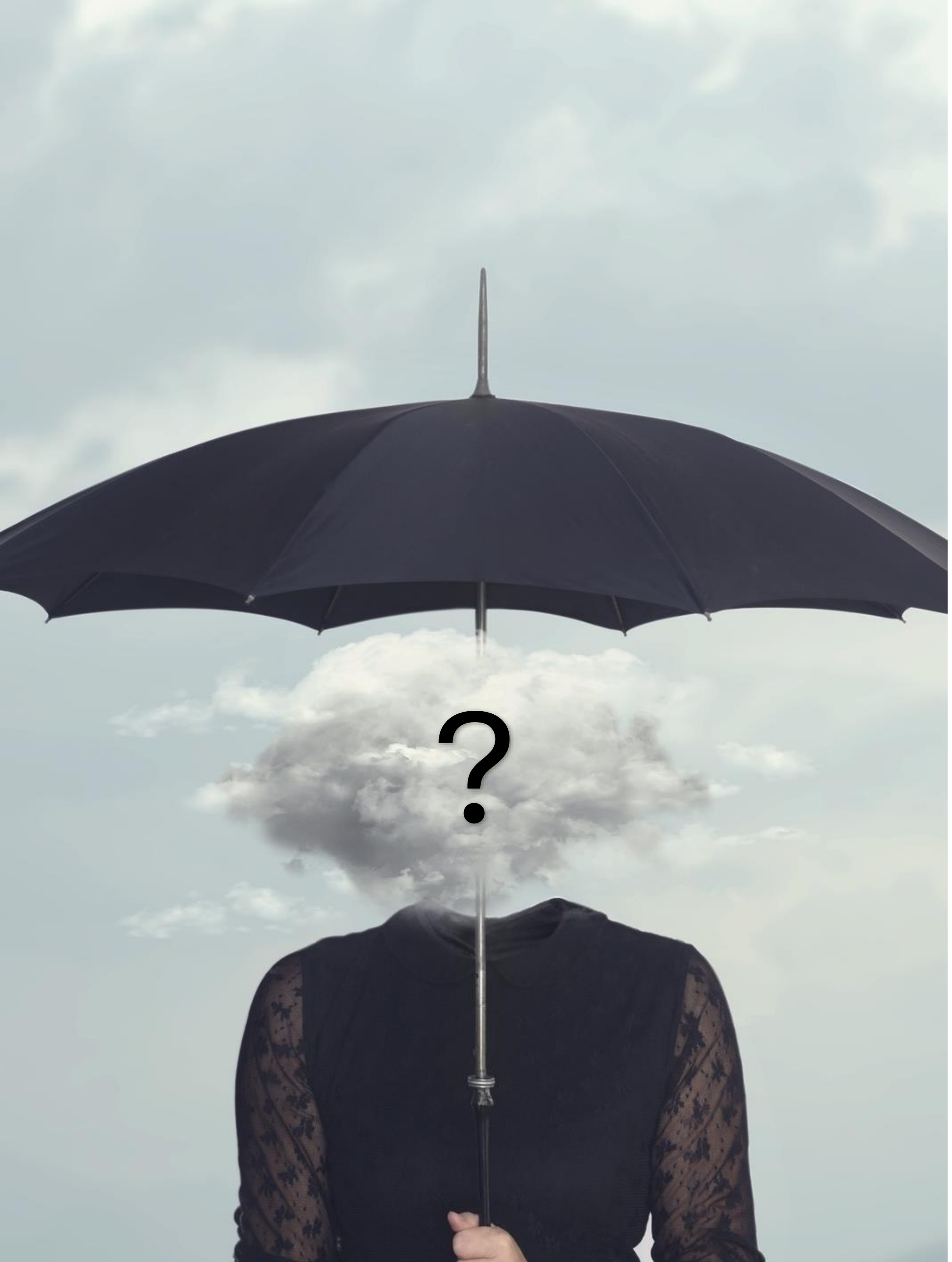
May 19, 2026



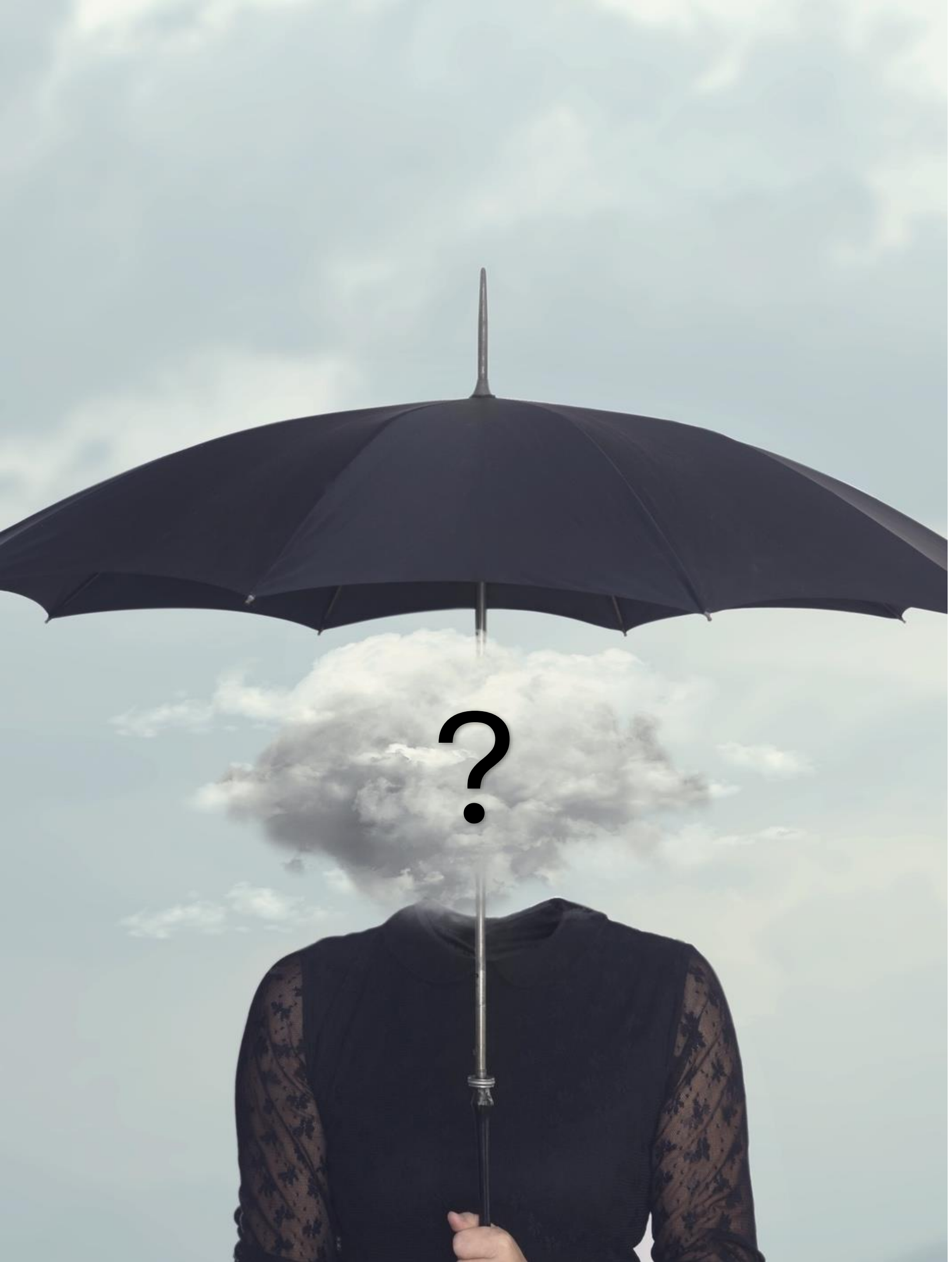
Where to start

What environmental data do you need to answer your research questions?

- *In what format do you need this data? (GeoTIFF? CSV?)*
- *Time frames?*
- *Locations?*

Do you need other spatial data? *(shape files)*

What other data might you be able to obtain from GEE?



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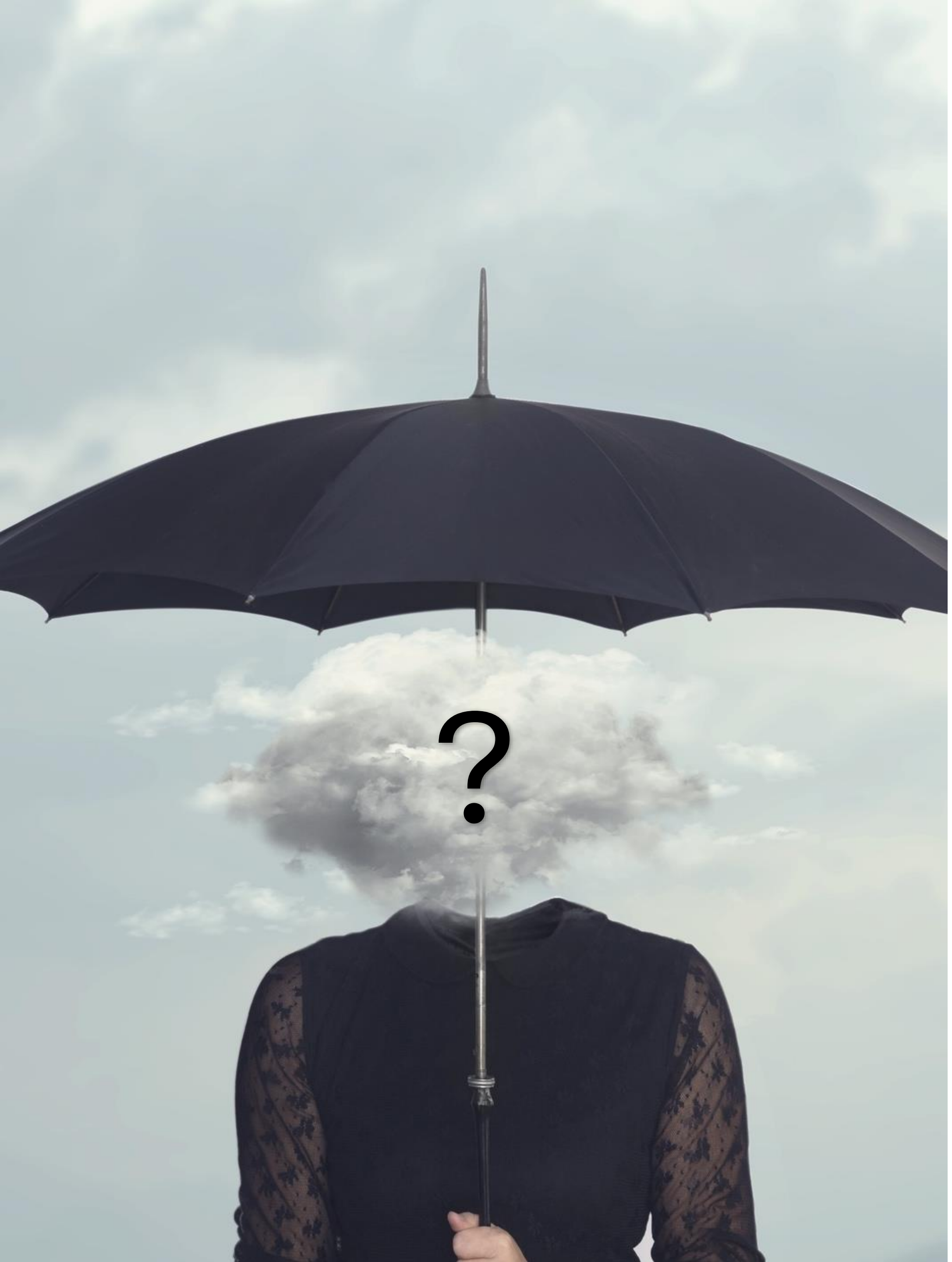

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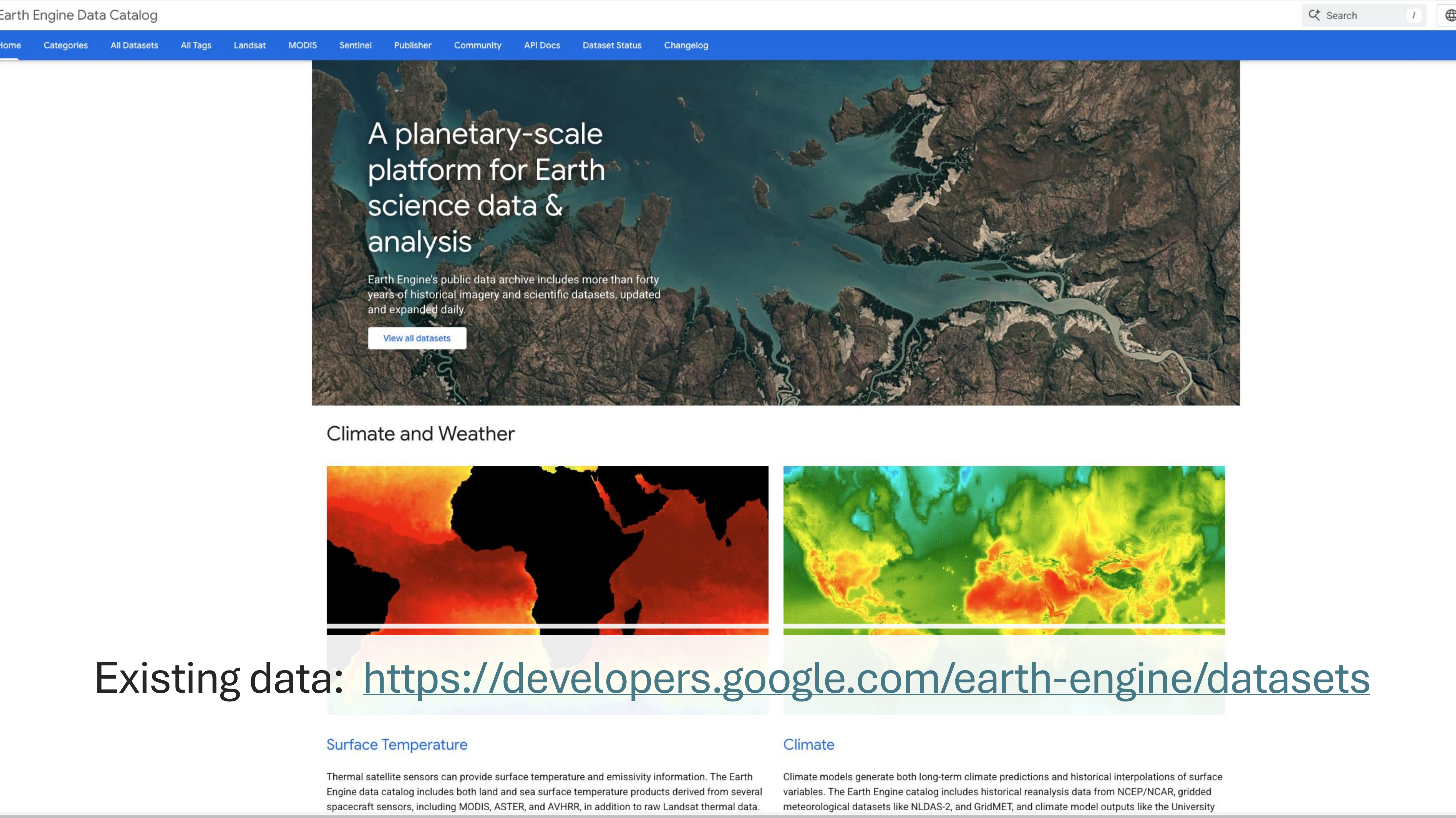


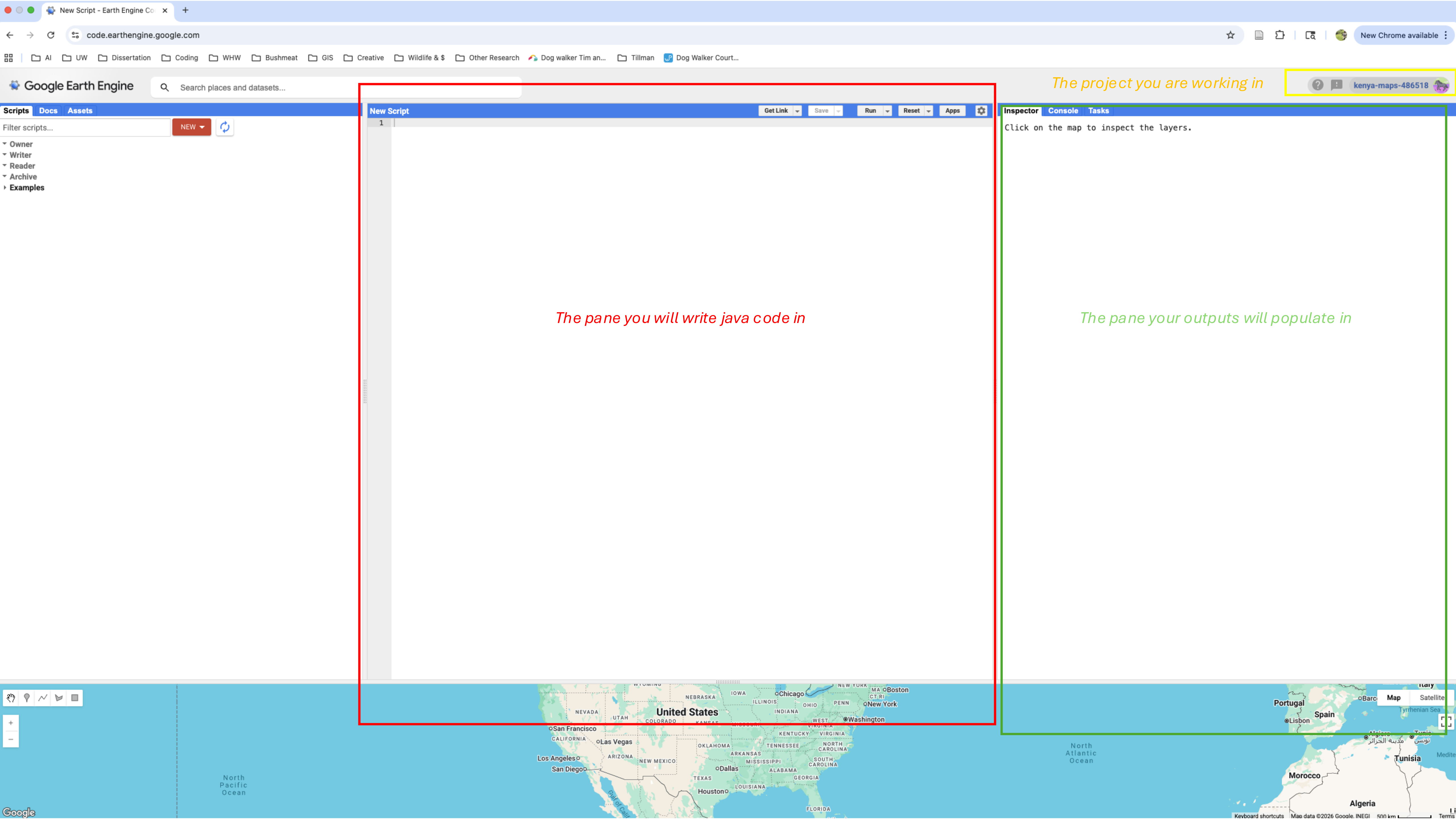
Existing data: https://developers.google.com/earth-engine/datasets

Let’s work through an example

- **Research question:** *What environmental factors influence wildlife and domestic animal movement across Kenyan landscapes, and how do these factors increase contact opportunities relevant to disease transmission?*
  - How can I measure resource availability driving animal movement?
    - NDVI (Normalized Difference Vegetation Index)
    - Rainfall
  - Years: 2017-2025
  - Study Site Locations: Samburu, Narok, Laikipia counties

Step 1: Setting up Google Earth Engine

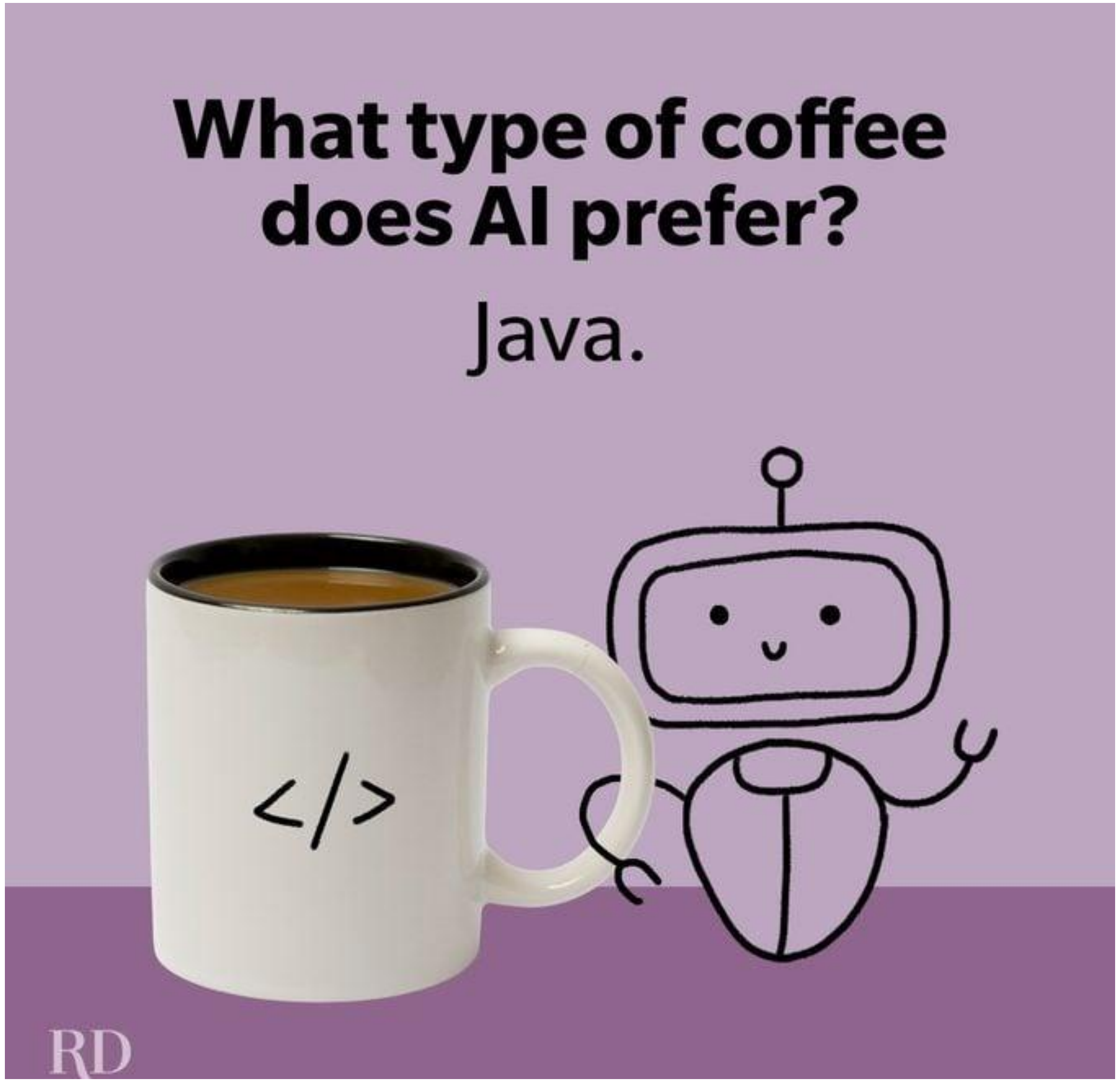
- You need a personal Gmail to set this up
- Follow these steps HERE to set up your GEE account:
    1. Go to the official Google Earth Engine homepage and click Get Started.
    2. Select Account: Log into your preferred Google/Gmail account.
    3. Register a Project: Earth Engine requires a linked Google Cloud project.
    4. **Select the Unpaid Usage option if you are a student**, researcher, or working in a non-profit/government sector Earth Engine Sign-up Flow Guide.
    5. Fill in Details: Enter your name, organization, and a brief description of how you intend to use the platform (e.g., "Academic remote sensing research") GIS Center Idaho State University Guide.
    6. Accept Terms: You will be routed to the Google Cloud Console to accept the Cloud Terms of Service.
    7. Confirmation: Wait for an approval email (usually within 24 hours).
- Once approved, login to your GEE account with your personal Gmail credentials



New Script - Earth Engine Co
code.earthengine.google.com
Google Earth Engine
Search places and datasets...
The project you are working in
kenya-maps-486518
Scripts
Docs
Assets
Filter scripts...
NEW
Owner
Writer
Reader
Archive
Examples
New Script
Get Link
Save
Run
Reset
Apps
Inspector
Console
Tasks
Click on the map to inspect the layers.
The pane you will write java code in
The pane your outputs will populate in
Map
Satellite
United States
North Pacific Ocean
North Atlantic Ocean

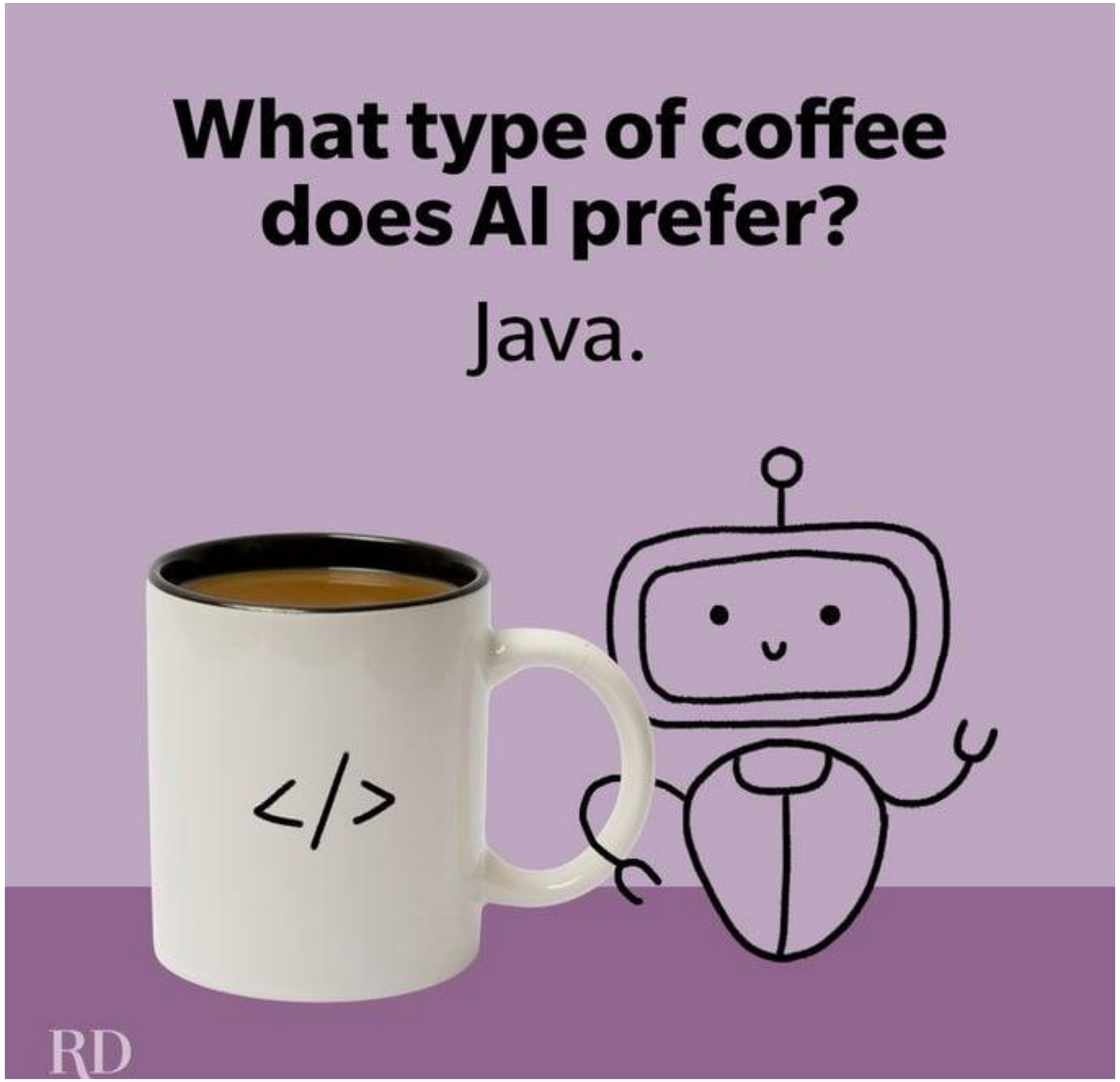
Using AI to code in Java in GEE

- Give your AI tool a very specific prompt.
- The more specific and detailed the better.
- AI is only as useful as you ask it to be.



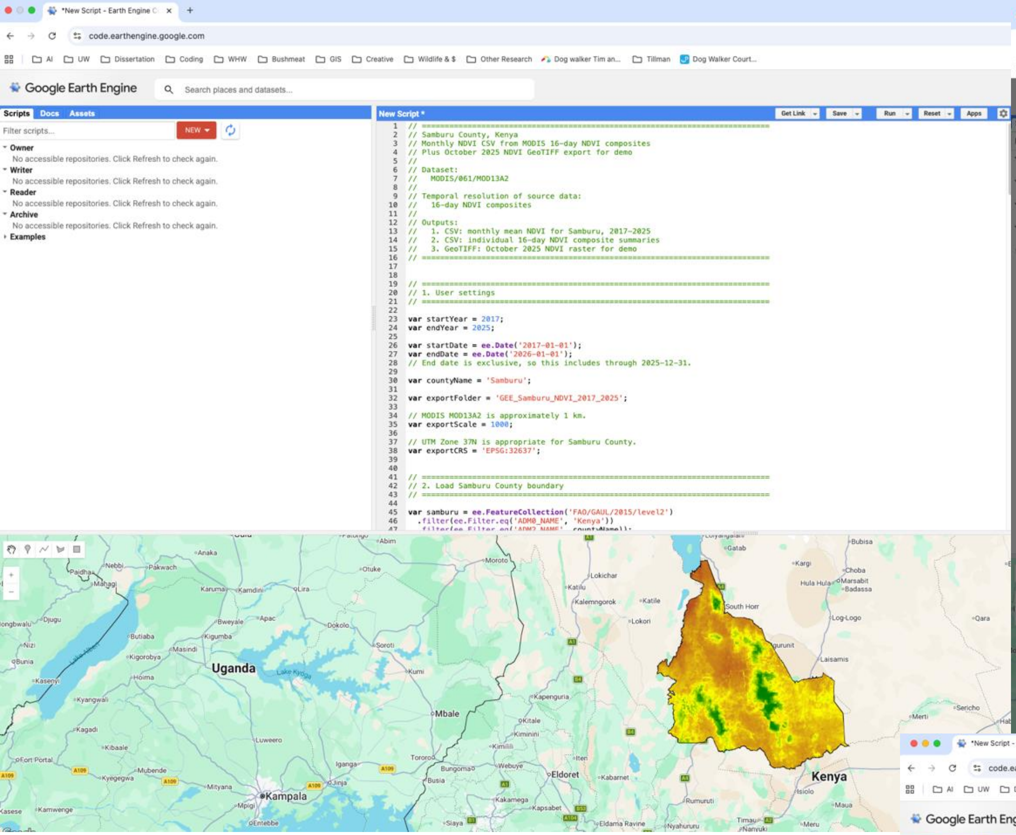
ChatGPT Prompt Example:

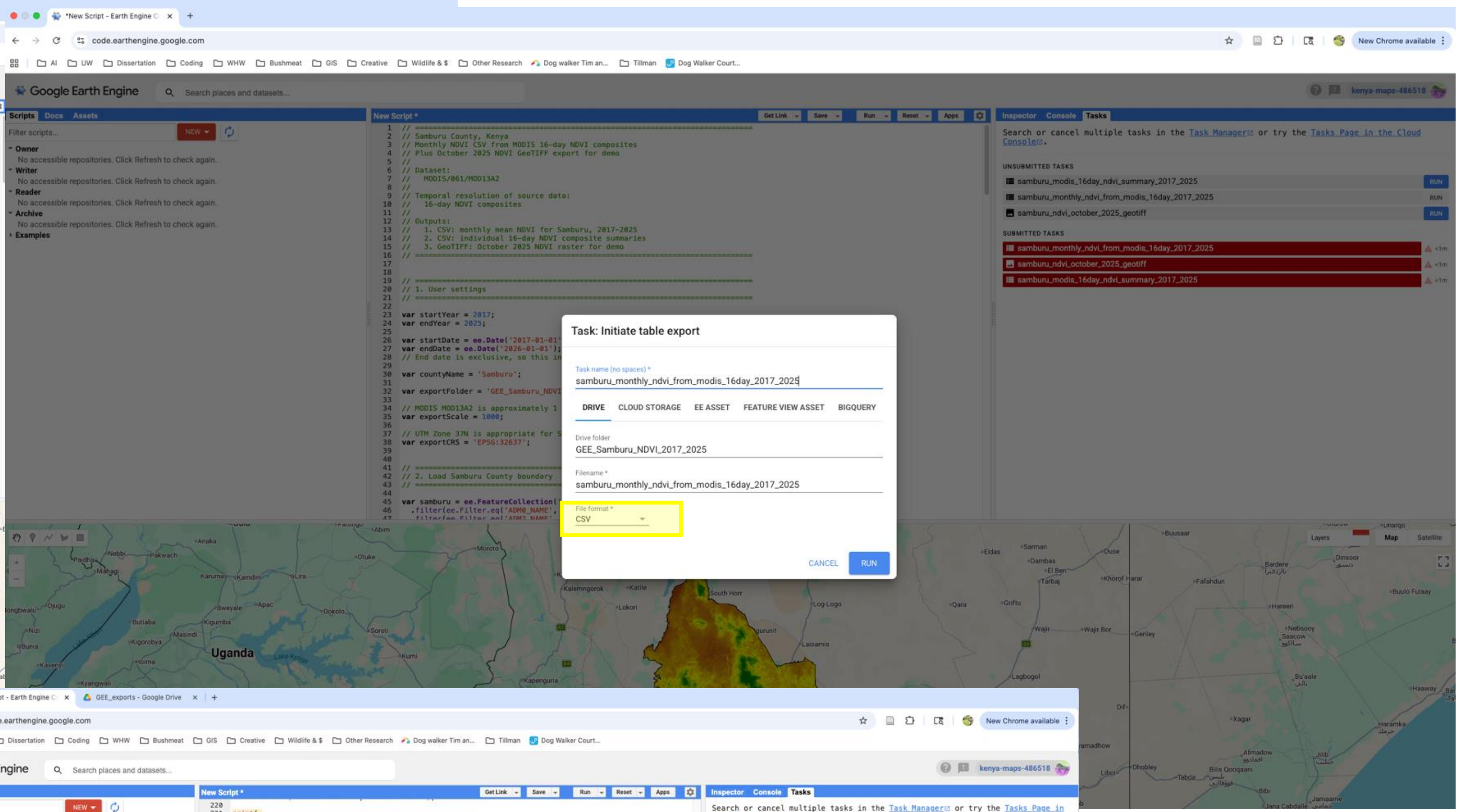
- ChatGPT prompt to recreate this code:
- Write a clean Google Earth Engine JavaScript script for Samburu County, Kenya. Use MODIS/061/MOD13A2 NDVI from 2017-01-01 through 2025-12-31.
- Load Samburu County from FAO/GAUL/2015/level2 using ADM0_NAME = Kenya and ADM2_NAME = Samburu.
- Apply the MODIS NDVI scale factor of 0.0001 and rename the band to ndvi.
- Do not clip each MODIS image inside the image collection map function, because that can cause MODIS projection transformation errors. Instead, simplify the Samburu geometry and use it only in reduceRegion(), Map.addLayer(), and Export.image.toDrive().
- The script should export three outputs to Google Drive:
    - 1. A CSV with one row per 16-day MODIS NDVI composite, giving mean NDVI over Samburu.
    - 2. A CSV with one row per month from 2017 to 2025, where monthly NDVI is calculated as the mean of available 16-day MODIS NDVI composites with start dates inside that calendar month.
    - 3. A GeoTIFF of mean NDVI for October 2025 for a live demo.
    - Add clear section headers and comments. Use export folder GEE_Samburu_NDVI_2017_2025, scale 1000, and CRS EPSG:32637. Make sure the script avoids the Date.get('day', 'month') timezone error by using date.get('day') only.

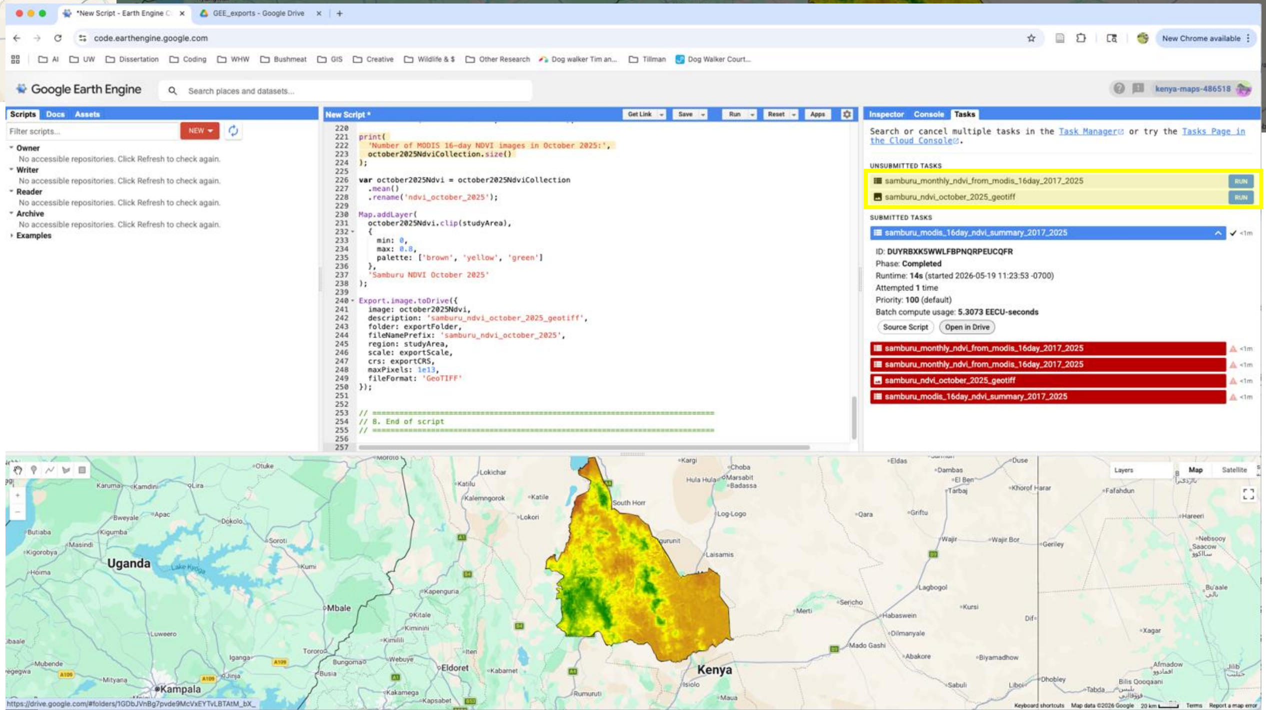


Step 3: Code in GEE

- Use your AI prompt to generate Java code and extract what you need from GEE


*New Script - Earth Engine
code.earthengine.google.com
Google Earth Engine
Search places and datasets...
Scripts Docs Assets
New Script *
Get Link Save Run Reset Apps
Owner
No accessible repositories. Click Refresh to check again.
Writer
No accessible repositories. Click Refresh to check again.
Reader
No accessible repositories. Click Refresh to check again.
Archive
No accessible repositories. Click Refresh to check again.
Examples
// Samburu County, Kenya
// Monthly NDVI CSV from MODIS 16-day NDVI composites
// Plus October 2025 NDVI GeoTIFF export for demo
// Dataset:
// MODIS/061/MOD13A2
// Temporal resolution of source data:
// 16-day NDVI composites
// Outputs:
// 1. CSV: monthly mean NDVI for Samburu, 2017-2025
// 2. CSV: individual 16-day NDVI composite summaries
// 3. GeoTIFF: October 2025 NDVI raster for demo
// 1. User settings
var startYear = 2017;
var endYear = 2025;
var startDate = ee.Date('2017-01-01');
var endDate = ee.Date('2026-01-01');
// End date is exclusive, so this includes through 2025-12-31.
var countyName = 'Samburu';
var exportFolder = 'GEE_Samburu_NDVI_2017_2025';
// MODIS MOD13A2 is approximately 1 km.
var exportScale = 1000;
// UTM Zone 37N is appropriate for Samburu County.
var exportCRS = 'EPSG:32637';
// 2. Load Samburu County boundary
var samburu = ee.FeatureCollection('FAO/GAUL/2015/level2')
.filter(ee.Filter.eq('ADM0_NAME', 'Kenya'))
Uganda
Kenya
Kampala


Inspector Console Tasks
Search or cancel multiple tasks in the Task Manager or try the Tasks Page in the Cloud Console.
UNSUBMITTED TASKS
samburu_modis_16day_ndvi_summary_2017_2025
samburu_monthly_ndvi_from_modis_16day_2017_2025
samburu_ndvi_october_2025_geotiff
SUBMITTED TASKS
samburu_monthly_ndvi_from_modis_16day_2017_2025
samburu_ndvi_october_2025_geotiff
samburu_modis_16day_ndvi_summary_2017_2025
Task: Initiate table export
Task name (no spaces) *
samburu_monthly_ndvi_from_modis_16day_2017_2025
DRIVE CLOUD STORAGE EE ASSET FEATURE VIEW ASSET BIGQUERY
Drive folder
GEE_Samburu_NDVI_2017_2025
Filename *
samburu_monthly_ndvi_from_modis_16day_2017_2025
File format *
CSV
CANCEL RUN


print(
'Number of MODIS 16-day NDVI images in October 2025:',
october2025NdviCollection.size()
);
var october2025Ndvi = october2025NdviCollection
.mean()
.rename('ndvi_october_2025');
Map.addLayer(
october2025Ndvi.clip(studyArea),
{
min: 0,
max: 0.8,
palette: ['brown', 'yellow', 'green']
},
'Samburu NDVI October 2025'
);
Export.image.toDrive({
image: october2025Ndvi,
description: 'samburu_ndvi_october_2025_geotiff',
folder: exportFolder,
fileNamePrefix: 'samburu_ndvi_october_2025',
region: studyArea,
scale: exportScale,
crs: exportCRS,
maxPixels: 1e13,
fileFormat: 'GeoTIFF'
});
// 8. End of script
UNSUBMITTED TASKS
samburu_monthly_ndvi_from_modis_16day_2017_2025
samburu_ndvi_october_2025_geotiff
SUBMITTED TASKS
samburu_modis_16day_ndvi_summary_2017_2025
ID: DUYRBXK5WWLFBPNQRPEUCQFR
Phase: Completed
Runtime: 14s (started 2026-05-19 11:23:53 -0700)
Attempted 1 time
Priority: 100 (default)
Batch compute usage: 5.3073 EECU-seconds
Source Script Open in Drive
samburu_monthly_ndvi_from_modis_16day_2017_2025
samburu_monthly_ndvi_from_modis_16day_2017_2025
samburu_ndvi_october_2025_geotiff
samburu_modis_16day_ndvi_summary_2017_2025
Uganda
Kenya
Kampala

Step 4: Import your GEE files into R

- Download your files from your folder in google drive
- Upload the CSVs, shapefiles, and geotiffs into your R project
- Happy coding!

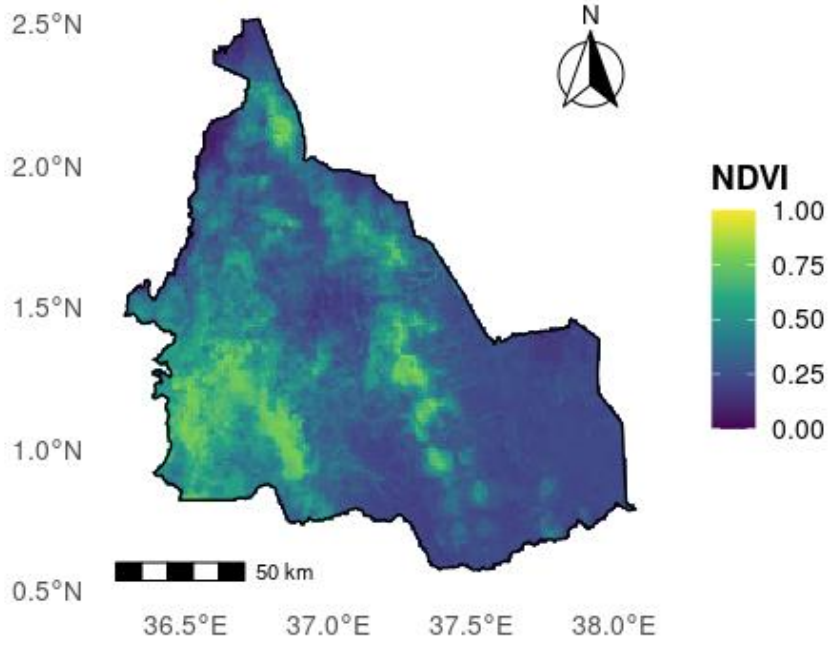
Example outputs in R

Monthly NDVI summary for Samburu County, Kenya, 2017-2025

Month	Season	16-day composites	Mean NDVI	SD	Minimum	Maximum
January	Hot dry season	18	0.356	0.081	0.273	0.532
February	Hot dry season	18	0.308	0.068	0.236	0.484
March	Long rains	18	0.320	0.073	0.230	0.461
April	Long rains	18	0.417	0.109	0.264	0.582
May	Long rains	18	0.451	0.092	0.333	0.629
June	Dry/cool season	18	0.381	0.064	0.292	0.494
July	Dry/cool season	18	0.338	0.048	0.262	0.419
August	Dry/cool season	18	0.335	0.055	0.262	0.443
September	Dry/cool season	18	0.322	0.043	0.271	0.404
October	Short rains	11	0.357	0.054	0.264	0.476
November	Short rains	16	0.407	0.084	0.286	0.549
December	Short rains	18	0.425	0.086	0.326	0.569

NDVI in Samburu County, Kenya

MODIS 16-day NDVI composites averaged for October 2025



NDVI: MODIS MOD13A2 via Google Earth Engine. Boundary: GADM via geodata.

A tibble: 12 × 8

month <dbl>	month_name <ord>	season <chr>	n_composites <int>	mean_ndvi <dbl>	sd_ndvi <dbl>	min_ndvi <dbl>	max_ndvi <dbl>
1	January	Hot dry season	18	0.356	0.081	0.273	0.532
2	February	Hot dry season	18	0.308	0.068	0.236	0.484
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12 rows

Figure 1: Average NDVI across study sites for October (2017–2025), shown as a representative snapshot of vegetation productivity during the domestic animal collaring period.

Example output in R, 2: Visualization of average NDVI for the month of October for all years of my study, across all study sites